UNIVERSITAS INDONESIA

Veritas, Probitas, Iustitia

FAKULTAS ILMU KOMPUTER

PENGANTAR STATISTIKA

Descriptive Statistics I: Statistical & Research Terms

Pengajar:

- Prof. Achmad Nizar Hidayanto
- Dr. Fatimah Azzahro
- Dr. Siti Aminah
- Dr. Nabila Clydea Harahap
- Larastri Kumaralalita, Ph.D.
- Kamila Alifia Imanuddin, M.T.I.

LEARNING OBJECTIVES

1. Know key statistical terms
2. Know key measurement terms
3. Know key research terms
4. Know the place of statistics in science

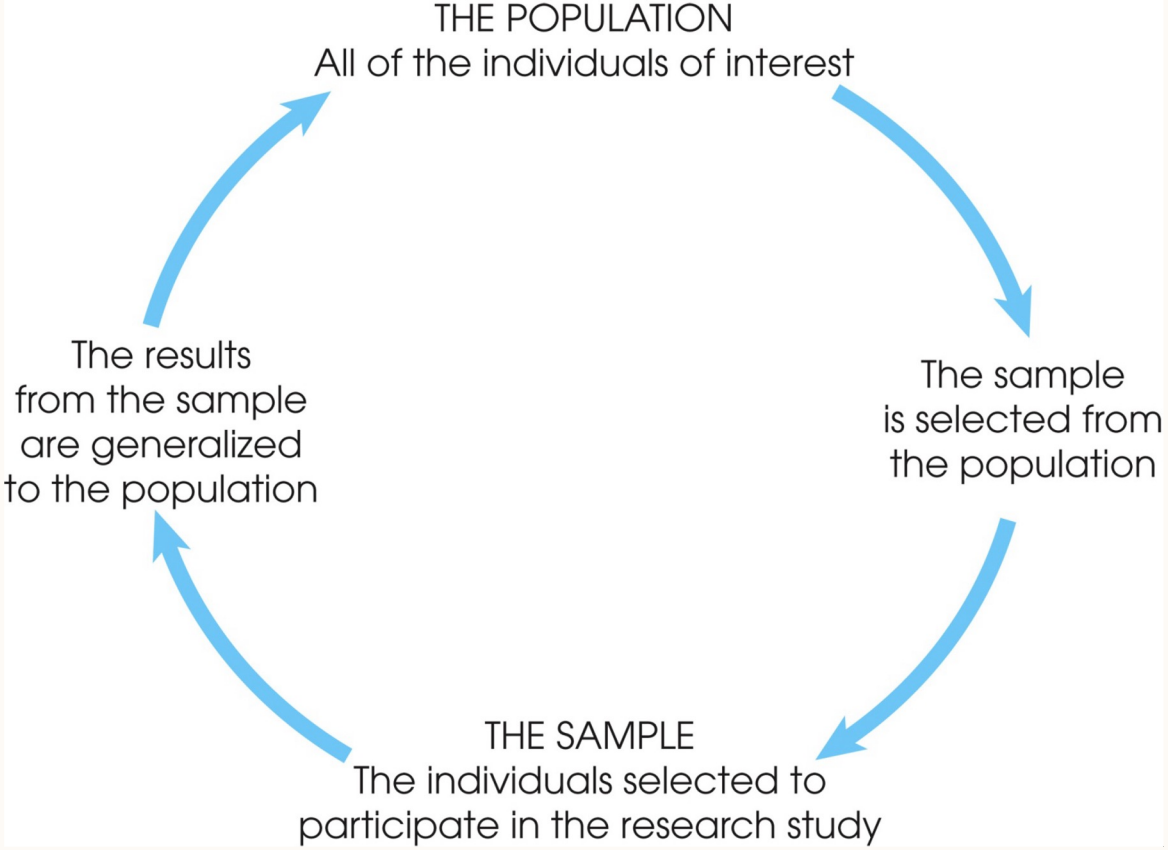
1.1 Statistics, Science and Observations

- "Statistics" means "*statistical procedures*"
- Uses of Statistics
  - Organize and summarize information
  - Determine exactly what conclusions are justified based on the results that were obtained
- Goals of statistical procedures
  - Accurate and meaningful interpretation
  - Provide standardized evaluation procedures

A. Populations and Samples

- Population
  - The set of all the individuals of interest in a particular study
  - Vary in size; often quite large
- Sample
  - A set of individuals selected from a population
  - Usually intended to represent the population in a research study

Relationship between population and sample



B. Variables and Data

- **Variable**
  - Characteristic or condition that changes or has different values for different individuals
- **Data (plural)**
  - Measurements or observations of a variable
- **Data set**
  - A collection of measurements or observations
- A **datum (singular)**
  - A single measurement or observation
  - Commonly called a **score** or **raw score**

C. Parameters and Statistics

Parameter

- A value, usually a numerical value, that describes a population
- Derived from measurements of the individuals in the population

Statistic

- A value, usually a numerical value, that describes a sample
- Derived from measurements of the individuals in the sample

D. Descriptive & Inferential Statistics

Descriptive statistics

- Summarize data
- Organize data
- Simplify data
- Familiar examples
  - Tables
  - Graphs
  - Averages

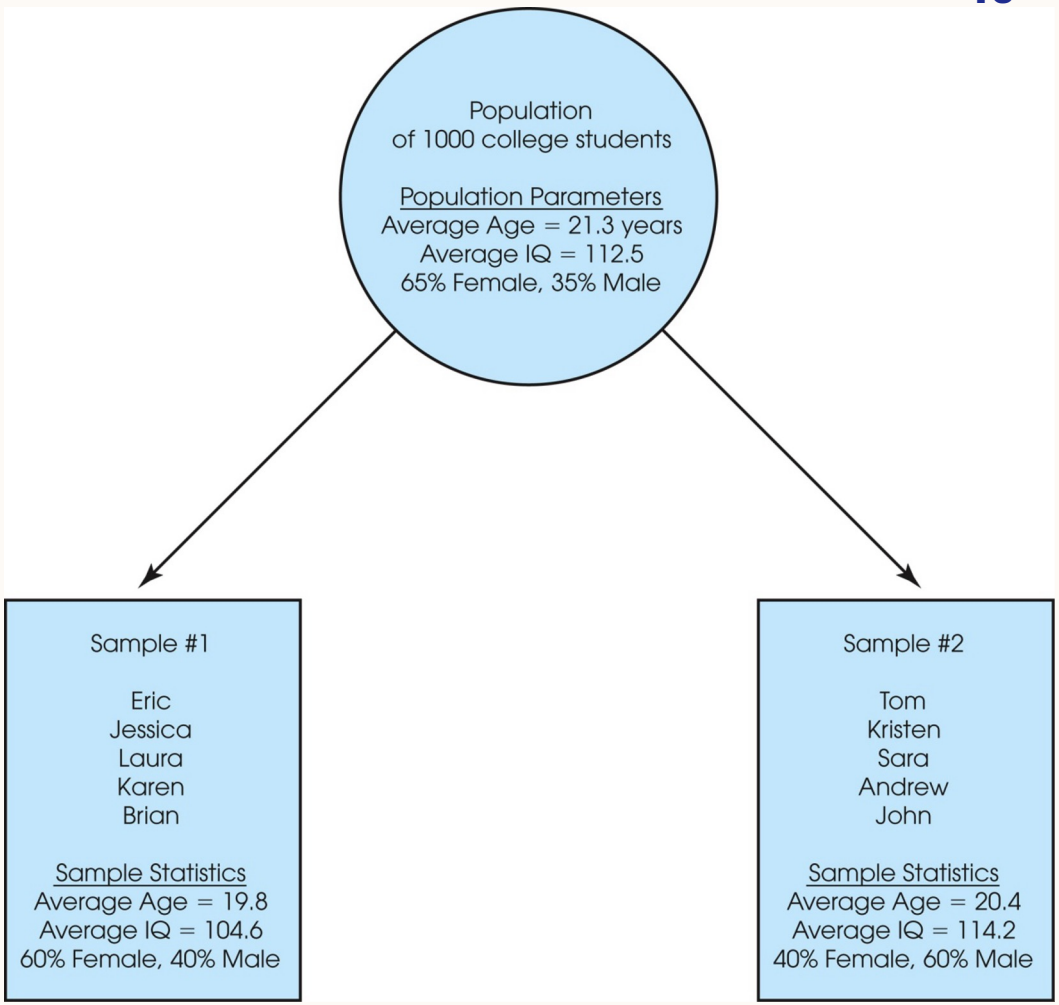
Inferential statistics

- Study samples to make generalizations about the population
- Interpret experimental data
- Common terminology
- "Margin of error"
- "Statistically significant"

E. Sampling Error

- Sample is never identical to population
- Sampling Error
  - The discrepancy, or amount of error, that exists between a sample statistic and the corresponding population parameter
- Example: Margin of Error in Polls
  - "This poll was taken from a sample of registered voters and has a margin of error of plus-or-minus 4 percentage points"

Figure 1.2
A demonstration of sampling error



F. Role of statistics in experimental research

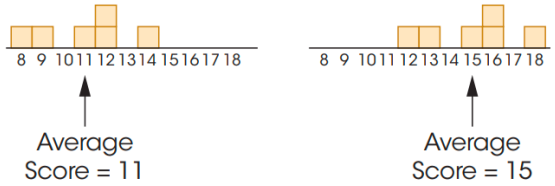
Step 1
Experiment:
Compare two studying methods

Population of College Students

Data:
Reading scores for the students in each sample

Sample A Read from computer screen	Sample B Read from printed pages
12 11 14	16 15 18
8 12 9	12 16 13

Step 2
Descriptive statistics:
Organize and simplify



Step 3
Inferential statistics:
Interpret the results

The sample data show a 4-point average difference between the two methods of studying. However, there are two ways to interpret the results.

1. There actually is no difference between the two studying methods, and the difference between the samples is due to chance (sampling error).
2. There really is a difference between the two methods of studying, and the sample data accurately reflect this difference.

The goal of inferential statistics is to help researchers decide between the two interpretations.

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Learning Check

A researcher is interested in the effect of amount of sleep on high school students' exam scores. A group of 75 high school boys agree to participate in the study. The boys are...

A. A statistic

B. A variable

C. A parameter

D. A sample

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A. A statistic

B. A variable

C. A parameter

D. **A sample**

Learning Check

- Decide if each of the following statements is **True** or **False**.

T/F

- Most research studies use data from samples

T/F

- When sample differs from the population there is a systematic difference between groups

Learning Check - Answer

True

- Samples used because it is not feasible or possible to measure all individuals in the population

False

- Sampling error due to random influences may produce unsystematic group differences

1.2 Observations, Variables and Measurement

- Scores are obtained by observing and measuring variables that scientists use to help define and explain external behaviors
- The process of measurement consists of applying carefully defined measurement procedures for each variable

A. Constructs & Operational Definitions

Constructs

- Internal attributes or characteristics that cannot be directly observed
- Useful for describing and explaining behavior

Operational Definition

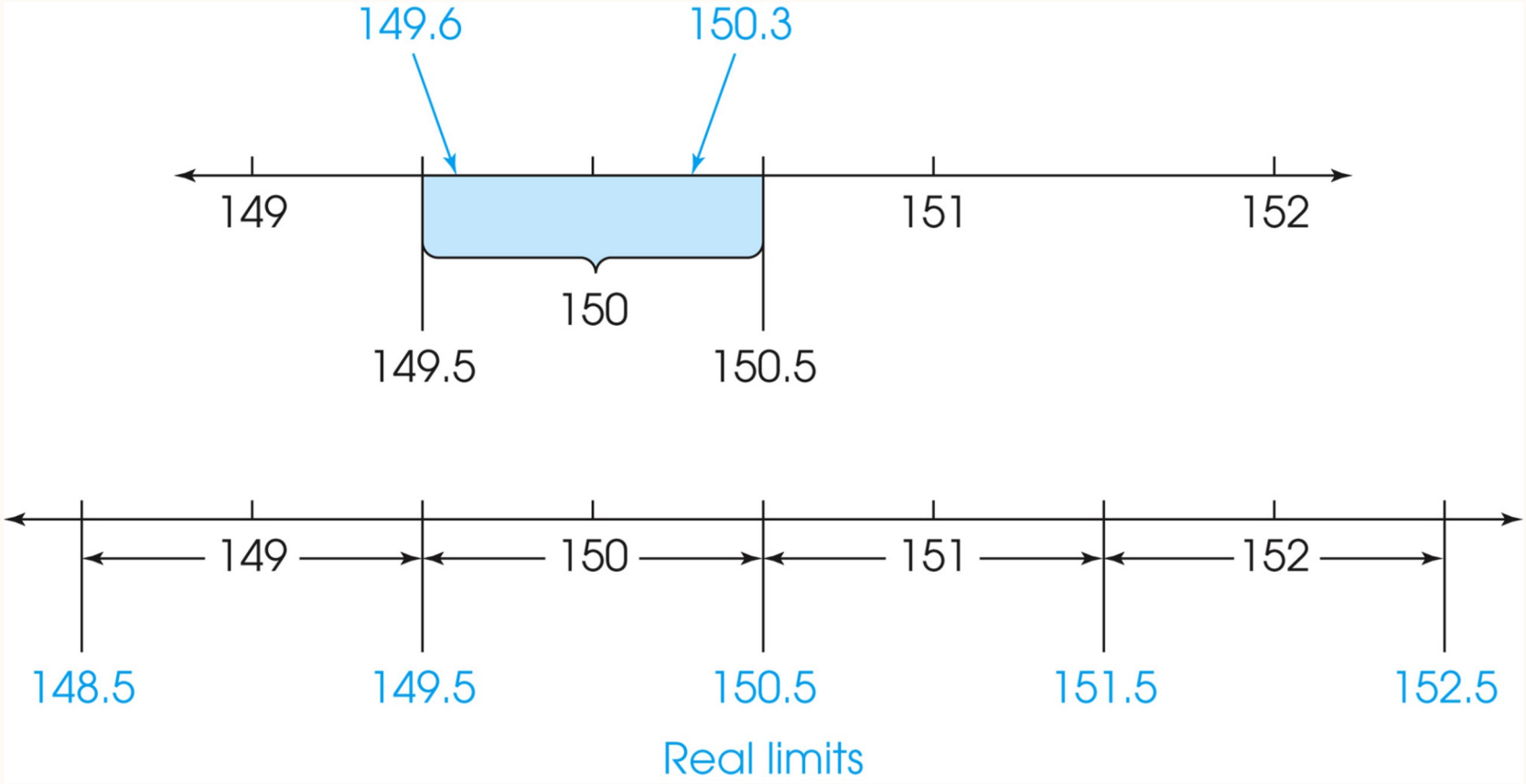
–Identifies the set of operations required to measure an external (observable) behavior

–Uses the resulting measurements as both a definition ***and*** a measurement of a hypothetical construct

B. Discrete and Continuous Variables

- **Discrete variable**
  - Has separate, indivisible categories
  - No values can exist between two neighboring categories
- **Continuous variable**
  - Have an infinite number of possible values between any two observed values
  - Every interval is divisible into an infinite number of equal parts

Example: Continuous Measurement



Real Limits of Continuous Variables

- **Real Limits** are the boundaries of each interval representing scores measured on a continuous number line
  - The real limit separating two adjacent scores is exactly halfway between the two scores
  - Each score has two real limits
    - The upper real limit marks the top of the interval
    - The lower real limit marks the bottom of the interval

C. Scales of Measurement

- Measurement assigns individuals or events to categories
  - The categories can simply be names such as male/female or employed/unemployed
  - They can be numerical values such as 68 inches or 175 pounds
- The complete set of categories makes up a scale of measurement
- Relationships between the categories determine different types of scales

Scales of Measurement

Scale	Characteristics	Examples
Nominal	• Label and categorize • No quantitative distinctions	• Gender • Diagnosis • Experimental or Control
Ordinal	• Categorizes observations • Categories organized by size or magnitude	• Rank in class • Clothing sizes (S,M,L,XL) • Olympic medals
Interval	• Ordered categories • Interval between categories of equal size • Arbitrary or absent zero point	• Temperature • IQ • Golf scores (above/below par)
Ratio	• Ordered categories • Equal interval between categories • Absolute zero point	• Number of correct answers • Time to complete task • Gain in height since last year

Scales of Measurement

Types of data on the basis of measurement

Scale	True Zero	Equal Intervals	Order	Category	Example
Nominal	No	No	No	Yes	Marital Status, Sex, Gender, Ethnicity
Ordinal	No	No	Yes	Yes	Student Letter Grade, NFL Team Rankings
Interval	No	Yes	Yes	Yes	Temperature in Fahrenheit, SAT Scores, IQ, Year
Ratio	Yes	Yes	Yes	Yes	Age, Height, Weight

Learning Check

- A study assesses the optimal size (number of other members) for study groups. The variable "Size of group" is ...

A • Discrete and interval

B • Continuous and ordinal

C • Discrete and ratio

D • Continuous and interval

Learning Check

- Decide if each of the following statements is **True** or **False**.

T/F

- Variables that cannot be measured directly cannot be studied scientifically

T/F

- Research measurements are made using specific procedures that define constructs

Learning Check - Answer

False	• Constructs (internal states) can only be ***observed*** indirectly, but can be operationally measured
True	• Operational definitions assure consistent measurement and provide construct definitions

1.3 Data Structures, Research Methods, and Statistics

- Individual Variables
  - A variable is observed
  - "Statistics" describe the observed variable
  - Category and/or numerical variables
- Relationships between variables
  - Two variables observed and measured
  - One of two possible data structures used to determine what type of relationship exists

Relationships Between Variables

A. Data Structure I: The Correlational Method

- One group of participants
- Measurement of two variables for each participant
- Goal is to describe type and magnitude of the relationship
- Patterns in the data reveal relationships
- Non-experimental method of study

Data structures for studies evaluating the relationship between variables

(a)

Student	Wake-up Time	Academic Performance
A	11	2.4
B	9	3.6
C	9	3.2
D	12	2.2
E	7	3.8
F	10	2.2
G	10	3.0
H	8	3.0

(b)

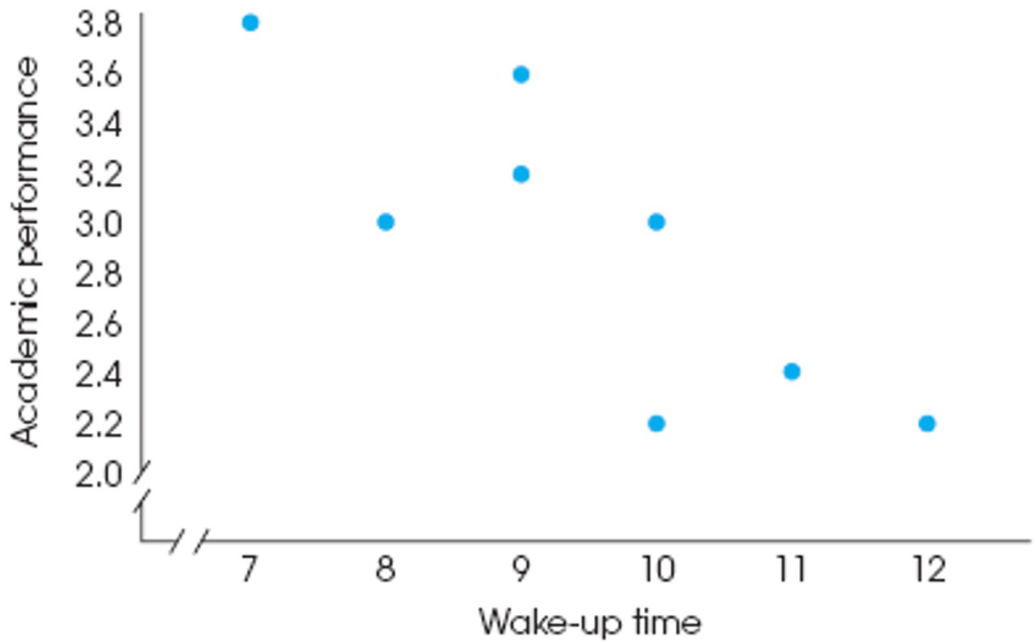


FIGURE 1.4

One of two data structures for studies evaluating the relationship between variables. Note that there are two separate measurements for each individual (wake-up time and academic performance). The same scores are shown in a table (a) and a graph (b).

Correlational Method Limitations

- Can demonstrate the existence of a relationship
- Does ***not*** provide an explanation for the relationship
- Most importantly, does ***not*** demonstrate a cause-and-effect relationship between the two variables

Relationships Between Variables

B. Data Structure II: Comparing two (or more) groups of Scores

- One variable defines the groups
- Scores are measured on second variable
- Both **experimental** and **non-experimental** studies use this structure

A. Experimental Method

- **Goal of Experimental Method**
  - To demonstrate a cause-and-effect relationship
- **Manipulation**
  - The level of one variable is determined by the experimenter
- **Control** rules out influence of other variables
  - Participant variables
  - Environmental variables

Data structure for studies comparing groups

One variable (violence/no violence) is used to define groups

A second variable (aggressive behavior) is measured to obtain scores within each group

Violence	No Violence
4	0
2	2
0	1
1	3
3	0
2	0
4	1
1	1
3	1

Compare groups of scores

Independent/Dependent Variables

- **Independent Variable** is the variable manipulated by the researcher
  - Independent because no other variable in the study influences its value
- **Dependent Variable** is the one observed to assess the effect of treatment
  - Dependent because its value is thought to depend on the value of the independent variable

Experimental Method: Control

- **Methods of control**
  - Random assignment of subjects
  - Matching of subjects
  - Holding level of some potentially influential variables constant
- **Control condition**
  - Individuals do not receive the experimental treatment.
  - They either receive no treatment or they receive a neutral, placebo treatment
  - Purpose: to provide a baseline for comparison with the experimental condition
- **Experimental condition**
  - Individuals do receive the experimental treatment

B. Non-experimental Methods

- Non-equivalent Groups
  - Researcher compares groups
  - Researcher cannot control who goes into which group
- Pre-test / Post-test
  - Individuals measured at two points in time
  - Researcher cannot control influence of the passage of time
- Independent variable is **quasi-independent**

Two examples of non-experimental studies

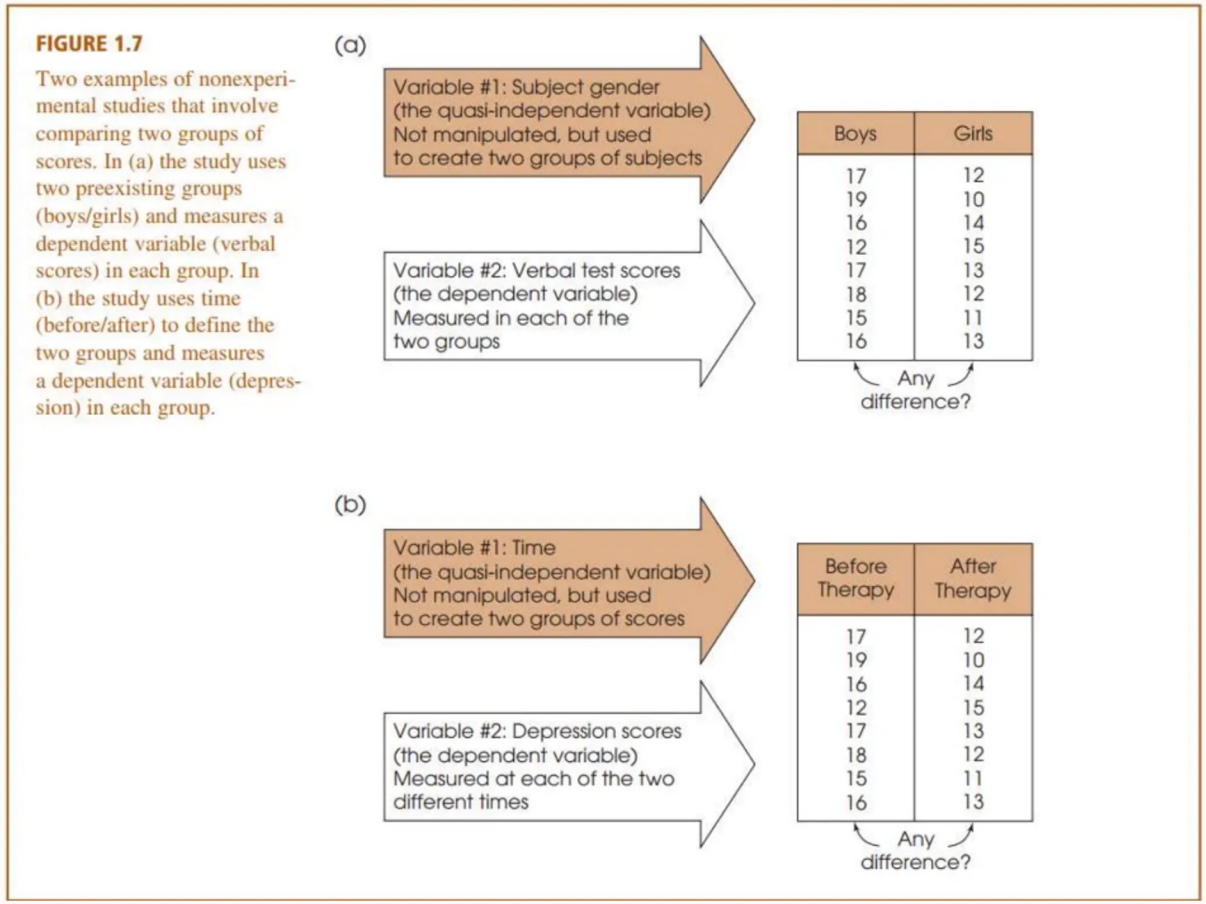


FIGURE 1.7

Two examples of nonexperimental studies that involve comparing two groups of scores. In (a) the study uses two preexisting groups (boys/girls) and measures a dependent variable (verbal scores) in each group. In (b) the study uses time (before/after) to define the two groups and measures a dependent variable (depression) in each group.

Learning Check

- Researchers observed that students exam scores were higher the more sleep they had the night before. This study is …

A • Descriptive

B • Experimental comparison of groups

C • Non-experimental group comparison

D • Correlational

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A • Descriptive

B • Experimental comparison of groups

C • Non-experimental group comparison

D • **Correlational**

Learning Check

- Decide if each of the following statements is **True** or **False**.

T/F

- All research methods have an independent variable

T/F

- All research methods can show cause-and-effect relationships

Learning Check - Answer

False

- Correlational methods do not need an independent variable

False

- Only experiments control the influence of participants and environmental variables

THANK YOU